

TD2 Answer Sheet

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Group:IF1

```
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ python src/main.py  
=== MarketPulse ===
```

```
Instrument  
AAPL - Apple Inc.  
Last price: 266.20 USD
```

```
Benchmark  
SP500 - S&P 500  
Last level: 6742.00
```

```
Period: 1 month  
Interval: Daily
```

```
Observations  
AAPL: 21  
SP500: 21
```

```
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ cat data/sample/instruments.json  
{  
  "instrument": {  
    "ticker": "AAPL",  
    "name": "Apple Inc.",  
    "currency": "USD",  
    "market": "NASDAQ"  
  },  
  "benchmark": {  
    "ticker": "SP500",  
    "name": "S&P 500",  
    "currency": "USD",  
    "market": "US"  
  }  
}
```

```
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ head data/sample/prices.csv  
date,ticker,open,high,low,close,volume  
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000  
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0  
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000  
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0  
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000  
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0  
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000  
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
```

```

@eliasbejj →/workspaces/esilv-marketpulse-g01-HANetteam (elias_bejjani) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANetteam (elias_bejjani) $

```

```

@eliasbejj →/workspaces/esilv-marketpulse-g01-HANetteam (elias_bejjani) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
250.00
<class 'str'>

```

Part 5-6-7-8) (final result)

```

@eliasbejj →/workspaces/esilv-marketpulse-g01-HANetteam (elias_bejjani) $ python src/main.py
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00

```

Part 5-6)

```
def filter_prices(prices, ticker):  
    return [row for row in prices if row["ticker"] == ticker]  
  
def get_first_close(prices):  
    return float(prices[0]["close"])  
  
def get_last_close(prices):  
    return float(prices[-1]["close"])  
  
def display_market_summary(asset, prices, show_currency=True):  
    print(f"{asset['ticker']} - {asset['name']}")  
    print(f"Observations : {len(prices)}")  
  
    first_close = get_first_close(prices)  
    last_close = get_last_close(prices)  
  
    if show_currency and "currency" in asset:  
        currency_label = f" {asset['currency']}"  
    else:  
        currency_label = ""  
  
    print(f"First close   : {first_close:.2f}{currency_label}")  
    print(f>Last close    : {last_close:.2f}{currency_label}")  
    print()
```